

1 Complexity Analysis

I Big O, little o, theta, big Omega, and small omega notation

i Definition

For real-valued nets f and g ,

- $\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M |g(x)|$ iff $f(x) \in O(g(x))$ and $|g(x)|$ is an asymptotic upper bound of $|f(x)|$.
- $\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \geq M |g(x)|$ iff $f(x) \in \Omega(g(x))$ and $|g(x)|$ is an asymptotic lower bound of $|f(x)|$.
- $\exists M_1, M_2 > 0, T \in \mathbb{R} : x \geq T \implies M_1 |g(x)| \leq |f(x)| \leq M_2 |g(x)|$ iff $f(x) \in \Theta(g(x))$.
- $\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| < M |g(x)|$ iff $f(x) \in o(g(x))$.
- $\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| > M |g(x)|$ iff $f(x) \in \omega(g(x))$.

$\in$ is also denoted as $=$ for them.

ii Limit

If $\exists k \in \mathbb{R} : x \geq k \implies g(x) \neq 0$,

- $\limsup \left| \frac{f(x)}{g(x)} \right| < \infty$ iff $f(x) \in O(g(x))$.

Proof.

$$|f(x)| \leq M |g(x)|, \quad x \geq T$$

$$\left| \frac{f(x)}{g(x)} \right| \leq M, \quad x \geq T$$

$$\sup \left| \frac{f(x)}{g(x)} \right| \leq M, \quad x \geq T$$

$$\limsup \left| \frac{f(x)}{g(x)} \right| \leq M < \infty$$

□

- $\liminf \left| \frac{f(x)}{g(x)} \right| > 0$ iff $f(x) \in \Omega(g(x))$.

Proof.

$$|f(x)| \geq M |g(x)|, \quad x \geq T$$

$$\left| \frac{f(x)}{g(x)} \right| \geq M, \quad x \geq T$$

$$\inf \left| \frac{f(x)}{g(x)} \right| \geq M, \quad x \geq T$$

$$\liminf \left| \frac{f(x)}{g(x)} \right| \geq M > 0$$

□

- $\infty > \limsup \left| \frac{f(x)}{g(x)} \right| \geq \liminf \left| \frac{f(x)}{g(x)} \right| > 0$ iff $f(x) \in \Theta(g(x))$.

Proof.

$$|f(x)| \geq M_1 |g(x)|, \quad x \geq T$$

$$\left| \frac{f(x)}{g(x)} \right| \geq M_1, \quad x \geq T$$

$$\inf \left| \frac{f(x)}{g(x)} \right| \geq M_1, \quad x \geq T$$

$$\liminf \left| \frac{f(x)}{g(x)} \right| \geq M_1 > 0$$

$$|f(x)| \leq M_2 |g(x)|, \quad x \geq T$$

$$\left| \frac{f(x)}{g(x)} \right| \leq M_2, \quad x \geq T$$

$$\sup \left| \frac{f(x)}{g(x)} \right| \leq M_2, \quad x \geq T$$

$$\limsup \left| \frac{f(x)}{g(x)} \right| \leq M_2 < \infty$$

□

$$\bullet \lim_{x \rightarrow \infty} \left| \frac{f(x)}{g(x)} \right| = 0 \text{ iff } f(x) \in o(g(x)).$$

$$|f(x)| < M |g(x)|$$

$$\left| \frac{f(x)}{g(x)} \right| < M$$

$$\lim_{x \rightarrow \infty} \left| \frac{f(x)}{g(x)} \right| = 0$$

$$\bullet \lim_{x \rightarrow \infty} \left| \frac{f(x)}{g(x)} \right| = \infty \text{ iff } f(x) \in \omega(g(x)).$$

$$|f(x)| > M |g(x)|$$

$$\left| \frac{f(x)}{g(x)} \right| > M$$

$$\lim_{x \rightarrow \infty} \left| \frac{f(x)}{g(x)} \right| = \infty$$

iii Relation

For any fixed $g(x)$, $o(g(x))$, $\Theta(g(x))$, and $\omega(g(x))$ are mutually exclusive.

Proof. Prove by contradiction.

If $f \in o(g(x)) \cap \Theta(g(x))$,

$$\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| < M |g(x)|$$

$$\exists M_1, M_2 > 0, U \in \mathbb{R} : x \geq U \implies M_1 |g(x)| \leq |f(x)| \leq M_2 |g(x)|$$

then set $M = M_1$, for $x \geq \max\{T, U\}$

$$M_1 |g(x)| \leq |f(x)| < M_1 |g(x)|,$$

which can't happen.

If $f \in \omega(g(x)) \cap \Theta(g(x))$,

$$\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| > M |g(x)|$$

$$\exists M_1, M_2 > 0, U \in \mathbb{R} : x \geq U \implies M_1 |g(x)| \leq |f(x)| \leq M_2 |g(x)|$$

then set $M = M_2$, for $x \geq \max\{T, U\}$

$$M_2 |g(x)| \geq |f(x)| > M_2 |g(x)|,$$

which can't happen.

If $f \in o(g(x)) \cap \omega(g(x))$,

$$\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| < M |g(x)|$$

$$\forall M > 0 : \exists U \in \mathbb{R} : x \geq U \implies |f(x)| > M |g(x)|$$

then set any fixed M ,

$$M |g(x)| < |f(x)| < M |g(x)|,$$

which can't happen.

□

For any fixed $g(x)$,

$$o(g(x)) \subseteq O(g(x))$$

Proof. Peak any $M > 0$.

□

For any fixed $g(x)$,

$$\omega(g(x)) \subseteq \Omega(g(x))$$

Proof. Peak any $M > 0$. □

For any fixed $g(x)$,

$$\Theta(g(x)) = O(g(x)) \cap \Omega(g(x))$$

Proof. Prove

$$\Theta(g(x)) \subseteq O(g(x)) \cap \Omega(g(x))$$

$$\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M |g(x)|$$

$$\exists N > 0, U \in \mathbb{R} : x \geq U \implies |f(x)| \geq N |g(x)|$$

then

$$\exists M, N > 0, \max\{T, U\} \in \mathbb{R} : x \geq \max\{T, U\} \implies N |g(x)| \leq |f(x)| \leq M |g(x)|$$

Prove

$$\Theta(g(x)) \supseteq O(g(x)) \cap \Omega(g(x))$$

$$\exists M_1, M_2 > 0, T \in \mathbb{R} : x \geq T \implies M_1 |g(x)| \leq |f(x)| \leq M_2 |g(x)|$$

then

$$\exists M_2 > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M_2 |g(x)|$$

$$\exists M_1 > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \geq M_1 |g(x)|$$

□

For any fixed $g(x)$,

$$O(g(x)) \setminus (o(g(x)) \cup \Theta(g(x))) \neq \emptyset$$

Proof. Counterexample:

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

$$g(x) = 1$$

$f \in O(g(x))$:

$$\exists 1 > 0, 1 \in \mathbb{R} : x \geq 1 \implies |f(x)| \leq |g(x)|$$

$f \notin o(g(x))$:

$$\forall 1 > 0, T \in \mathbb{R} : \exists x \geq T, x \in \mathbb{Q} : |f(x)| = |g(x)|$$

$f \notin \Theta(g(x))$:

$$\forall M_1 > 0, x \in \mathbb{R}, x \notin \mathbb{Q} : M_1 |g(x)| = M_1 > |f(x)| = 0$$

□

For any fixed $g(x)$,

$$\Omega(g(x)) \setminus (\omega(g(x)) \cup \Theta(g(x))) \neq \emptyset$$

Proof. Counterexample:

$$f(x) = 1$$

$$g(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

$f \in \Omega(g(x))$:

$$\exists 1 > 0, 1 \in \mathbb{R} : x \geq 1 \implies |f(x)| \geq |g(x)|$$

$f \notin \omega(g(x))$:

$$\forall 1 > 0, T \in \mathbb{R} : \exists x \geq T, x \in \mathbb{Q} : |f(x)| = |g(x)|$$

$f \notin \Theta(g(x))$:

$$\forall M_2 > 0, x \in \mathbb{R}, x \notin \mathbb{Q} : M_2 |g(x)| = 0 < |f(x)| = 1$$

□

$$f(x) \in O(g(x)) \iff g(x) \in \Omega(f(x))$$

Proof. $\implies$:

$$\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M |g(x)|$$

then

$$\exists \frac{1}{M} > 0, T \in \mathbb{R} : x \geq T \implies |g(x)| \geq \frac{1}{M} |f(x)|$$

$\Leftarrow$:

$$\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \geq M |g(x)|$$

then

$$\exists \frac{1}{M} > 0, T \in \mathbb{R} : x \geq T \implies |g(x)| \leq \frac{1}{M} |f(x)|$$

□

$$f(x) \in o(g(x)) \iff g(x) \in \omega(f(x))$$

Proof. $\implies$:

$$\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| < M |g(x)|$$

then

$$\forall \frac{1}{M} > 0 : \exists T \in \mathbb{R} : x \geq T \implies |g(x)| > \frac{1}{M} |f(x)|$$

$\Leftarrow$:

$$\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| > M |g(x)|$$

then

$$\forall \frac{1}{M} > 0 : \exists T \in \mathbb{R} : x \geq T \implies |g(x)| < \frac{1}{M} |f(x)|$$

□

For any fixed $g(x)$,

$$\exists f(x) : f(x) \notin O(g(x)) \cup \Omega(g(x))$$

Proof.

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ x, & x \notin \mathbb{Q} \end{cases}$$

$$g(x) = \begin{cases} x, & x \in \mathbb{Q} \\ 1, & x \notin \mathbb{Q} \end{cases}$$

For $x \in \mathbb{R} \setminus \mathbb{Q}$,

$$\nexists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M |g(x)|$$

For $x \in \mathbb{Q}$,

$$\nexists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \geq M |g(x)|$$

□

iv Binary relation

- $\Theta(g(x))$ is an equivalence class.

Proof. Transitive:

$$\exists M_1, M_2 > 0, T \in \mathbb{R} : x \geq T \implies M_1 |g(x)| \leq |f(x)| \leq M_2 |g(x)|$$

$$\exists N_1, N_2 > 0, U \in \mathbb{R} : x \geq U \implies N_1 |h(x)| \leq |g(x)| \leq N_2 |h(x)|$$

then

$$\exists M_1 N_1, M_2 N_2 > 0, \max\{T, U\} \in \mathbb{R} : x \geq \max\{T, U\} \implies M_1 N_1 |h(x)| \leq |f(x)| \leq M_2 N_2 |h(x)|$$

Reflexive:

$$f(x) \in \Theta(f(x))$$

$$\exists 1, 1 > 0, a \in \mathbb{R} : x \in \mathbb{R} \implies 1 |f(x)| \leq |f(x)| \leq 1 |f(x)|$$

Symmetric:

$$\exists M_1, M_2 > 0, T \in \mathbb{R} : x \geq T \implies M_1 |g(x)| \leq |f(x)| \leq M_2 |g(x)|$$

then

$$\exists \frac{1}{M_2}, \frac{1}{M_1} > 0, T \in \mathbb{R} : x \geq T \implies \frac{1}{M_2} |f(x)| \leq |g(x)| \leq \frac{1}{M_1} |f(x)|$$

□

- $O(g(x))$ is partial-ordered set with Θ as $=$.

Proof. Transitive:

$$\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M |g(x)|$$

$$\exists N > 0, U \in \mathbb{R} : x \geq U \implies |g(x)| \leq N |h(x)|$$

then

$$\exists MN > 0, \max\{T, U\} \in \mathbb{R} : x \geq \max\{T, U\} \implies |f(x)| \leq MN |h(x)|$$

Reflexive:

$$\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M |g(x)|$$

then

$$\exists \frac{1}{M} > 0, T \in \mathbb{R} : x \geq T \implies |g(x)| \leq \frac{1}{M} |f(x)|$$

Antisymmetric:

$$\exists M > 0, T \in \mathbb{R} : x \geq T \implies |f(x)| \leq M |g(x)|$$

$$\exists N > 0, U \in \mathbb{R} : x \geq U \implies |g(x)| \leq N |f(x)|$$

then

$$\exists M, \frac{1}{N} > 0, \max\{T, U\} \in \mathbb{R} : x \geq \max\{T, U\} \implies \frac{1}{N} |g(x)| \leq |f(x)| \leq M |g(x)|$$

□

- $\Omega(g(x))$ is partial-ordered set with Θ as $=$.

Proof. Similar to that of O .

□

- $o(g(x))$ is a strictly partially-ordered set.

Proof. Asymmetry:

$$f(x) \in o(g(x)) \iff g(x) \in \omega(f(x)) \implies g \notin o(f(x))$$

Transitivity:

$$\forall M > 0 : \exists T \in \mathbb{R} : x \geq T \implies |f(x)| < M |g(x)|$$

$$\exists U \in \mathbb{R} : x \geq U \implies |g(x)| < |h(x)|$$

then

$$\forall M > 0 : \exists \max\{T, U\} \in \mathbb{R} : x \geq \max\{T, U\} \implies |f(x)| < M |h(x)|$$

□

- $\omega(g(x))$ is a strictly partially-ordered set.

Proof. Similar to that of o .

□

II Master Theorem

i Integral test (for convergence) or Maclaurin–Cauchy test

Suppose $f(x) : [a, \infty) \rightarrow \mathbb{R}$ for some $a \in \mathbb{N}$ is monotone and $\lim_{x \rightarrow \infty} f(x) = 0$. Then $\int_a^\infty f(x) dx$ is convergent iff $\sum_{k=a}^\infty f(k)$ is convergent. And,

- If $\int_a^\infty f(x) dx$ is convergent and $f(x)$ is monotone decreasing, for any $n \in \mathbb{N}_{\geq a}$,

$$\int_n^\infty f(x) dx \leq \sum_{k=n}^\infty f(k) \leq f(n) + \int_n^\infty f(x) dx,$$

called remainder estimate for integral test. Note that for $n \in \mathbb{N}_{>a}$, $f(n) + \int_n^\infty f(x) dx \leq \int_{n-1}^\infty f(x) dx$.

- If $\int_a^\infty f(x) dx$ is convergent and $f(x)$ is monotone increasing, for any $n \in \mathbb{N}_{\geq a}$,

$$\int_n^\infty f(x) dx \geq \sum_{k=n}^\infty f(k) \geq f(n) + \int_n^\infty f(x) dx,$$

called remainder estimate for integral test. Note that for $n \in \mathbb{N}_{>a}$, $f(n) + \int_n^\infty f(x) dx \geq \int_{n-1}^\infty f(x) dx$.

Proof. Consider the case where $f(x)$ is monotone increasing. The case where $f(x)$ is monotone decreasing can be proven similarly.

First consider the case where $\int_a^\infty f(x) dx$ is convergent. This implies that $f(x)$ is non-positive. Thus, for any fixed $n \in \mathbb{N}_{\geq a}$,

$$\int_n^{n+1} f(x) dx \geq f(n),$$

and for any $k \geq n+1$,

$$\int_k^{k+1} f(x) dx \geq f(k) \geq \int_{k-1}^k f(x) dx.$$

For any $N \in \mathbb{N}_{\geq n}$, summing over all $k \in \mathbb{N}_{\geq n}$, we obtain,

$$\int_n^{N+1} f(x) dx \geq \sum_{k=n}^N f(k) \geq f(n) + \int_n^N f(x) dx.$$

Take limit $\lim_{N \rightarrow \infty}$, we obtain

$$\int_n^\infty f(x) dx \geq \sum_{k=n}^\infty f(k) \geq f(n) + \int_n^\infty f(x) dx.$$

Second, consider the case where $\int_a^\infty f(x) dx$ is divergent. Since $f(x)$ is monotone. Either $\int_a^\infty f(x) dx = -\infty$ or $\int_a^\infty f(x) dx = \infty$. For the former, the supremum of $f(x)$ is non-positive, and $\sum_{k=a}^\infty f(k) = -\infty$. For the latter, there exists $x_0 > a$ such that $f(x) > 0$ for all $x \geq x_0$, and thus $\sum_{k=a}^\infty f(k) = \infty$. □

ii p-test (for improper integral)

For any fixed $a > 0$, $\int_a^\infty \frac{1}{x^p} dx$ converges iff $p > 1$, and if so,

$$\int_a^\infty \frac{1}{x^p} dx = \frac{1}{(p-1)a^{p-1}};$$

$\int_0^a \frac{1}{x^p} dx$ converges iff $p < 1$, and if so,

$$\int_0^a \frac{1}{x^p} dx = -\frac{1}{(p-1)a^{p-1}}.$$

Proof.

$$\int \frac{1}{x^p} dx = \begin{cases} -\frac{1}{(p-1)x^{p-1}}, & p \neq 1 \\ \ln(x), & p = 1 \end{cases}$$

$$\forall p < 1 : \nexists \lim_{x \rightarrow \infty} -\frac{1}{(p-1)x^{p-1}} \\ \nexists \lim_{x \rightarrow \infty} \ln(x)$$

$$\forall p > 1 : \lim_{x \rightarrow \infty} -\frac{1}{(p-1)x^{p-1}} = 0 \quad \int_a^\infty \frac{1}{x^p} dx = \frac{1}{(p-1)a^{p-1}}$$

$$\forall p > 1 : \nexists \lim_{x \rightarrow 0^+} -\frac{1}{(p-1)x^{p-1}} \\ \nexists \lim_{x \rightarrow 0^+} \ln(x)$$

$$\forall p < 1 : \lim_{x \rightarrow 0^+} -\frac{1}{(p-1)x^{p-1}} = 0 \quad \int_0^a \frac{1}{x^p} dx = -\frac{1}{(p-1)a^{p-1}}$$

□

iii Generalized p-test (for improper integral)

Define iterated logarithmic function $L_n(x)$ as

$$L_0(x) = x, \quad x \in \mathbb{R}_{>0}$$

$$L_n(x) = \ln(L_{n-1}(x)), \quad n \in \mathbb{N}, x \in \mathbb{R}_{>0}.$$

For any fixed $a > 1, k \in \mathbb{N}$,

$$\int_a^\infty \frac{1}{\left(\prod_{i=0}^{k-1} L_i(x)\right) (L_k(x))^m} dx$$

converges iff $m > 1$.

Proof. Let

$$u = L_k(x)$$

$$du = \frac{dx}{\prod_{i=0}^{k-1} L_i(x)}$$

$$\int \frac{1}{\left(\prod_{i=0}^{k-1} L_i(x)\right) (L_k(x))^m} dx = \int \frac{1}{u^m} du$$

By p-test, the statement holds. □

For any fixed $0 < a < 1$,

$$\int_0^a \frac{1}{x(\ln x)^m} dx$$

converges iff $m < 1$.

Proof. By substitution of $u = \ln x$. □

iv Generalized p-test for series or generalized p-series test

Define iterated logarithmic function $L_n(x)$ as

$$L_0(x) = x, \quad x \in \mathbb{R}_{>0}$$

$$L_n(x) = \ln(L_{n-1}(x)), \quad n \in \mathbb{N}, x \in \mathbb{R}_{>0}.$$

For any fixed $a \in \mathbb{N}$,

$$\sum_{n=a}^\infty \frac{1}{\left(\prod_{i=0}^{k-1} L_i(n)\right) (L_k(n))^m}$$

converges iff $m > 1$.

Proof. By integral test and generalized p-test for improper integral, the statement holds. □

v Generalized Master Theorem

For a positive valued net $T(n)$ defined on $n = b^h$ for any $h \in \mathbb{N}_0$ as

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

for some constants $a > 0$ and $b > 1$, and non-negative-valued function $f(n)$,

1. If for some $c < \log_b a$,

$$f(n) = O(n^c),$$

then

$$T(n) = \Theta(n^{\log_b a}).$$

2. Define iterated logarithmic function $L_n(x)$ as

$$L_0(x) = x, \quad x \in \mathbb{R}_{>0}$$

$$L_n(x) = \ln(L_{n-1}(x)), \quad n \in \mathbb{N}, x \in \mathbb{R}_{>0}.$$

(a) If for some positive integers k and positive number m ,

$$f(n) = \Theta \left(n^{\log_b a} \left(\prod_{i=1}^{k-1} (L_i(n))^{-1} \right) (L_k(n))^{m-1} \right),$$

then

$$T(n) = \Theta \left(n^{\log_b a} (L_k(n))^m \right).$$

(b) If for some positive integer k ,

$$f(n) = \Theta \left(n^{\log_b a} \prod_{i=1}^k (L_i(n))^{-1} \right),$$

then

$$T(n) = \Theta \left(n^{\log_b a} L_{k-1}(n) \right).$$

(c) If for some positive integers k and some positive number m ,

$$f(n) = \Theta \left(n^{\log_b a} \left(\prod_{i=1}^{k-1} (L_i(n))^{-1} \right) (L_k(n))^{-m-1} \right),$$

then

$$T(n) = \Theta \left(n^{\log_b a} \right).$$

3. If $f(n) = \Omega(n^c)$ for some $c > \log_b a$, and the regularity condition that $\exists m \in \mathbb{R}, k < 1$ such that $n \geq m$ implies

$$af\left(\frac{n}{b}\right) \leq kf(n)$$

is satisfied, then

$$T(n) = \Theta(f(n)).$$

Proof. Let $s = \log_b a$, $n = b^h$, and $f(x) = x^s g(x)$. At level j , there are a^j nodes, each of size $\frac{n}{b^j}$. Thus, the total non-recursive work at level j is

$$a^j f\left(\frac{n}{b^j}\right).$$

Hence

$$\begin{aligned} T(n) &= \sum_{j=0}^{h-1} a^j f\left(\frac{n}{b^j}\right) + n^s T(1) \\ &= \sum_{j=0}^{h-1} b^{sj} f\left(\frac{b^h}{b^j}\right) + \Theta(n^s) \\ &= n^s \sum_{j=0}^{h-1} g(b^{h-j}) + \Theta(n^s) = n^s \sum_{r=1}^h g(b^r) + \Theta(n^s) \end{aligned}$$

Case 1: If for some $c < \log_b a$,

$$f(n) = O(n^c),$$

then

$$T(n) = \Theta \left(n^{\log_b a} \right).$$

Case 2:

$$\begin{aligned} L_k(b^x) &= \Theta(L_{k-1}(x)) \\ f(n) &= \Theta \left(n^{\log_b a} \left(\prod_{i=1}^{k-1} (L_i(n))^{-1} \right) (L_k(n))^{m-1} \right) \\ g(b^r) &= \Theta \left(\left(\prod_{i=1}^{k-2} (L_i(r))^{-1} \right) (L_{k-1}(r))^{m-1} \right) \end{aligned}$$

By generalized p-test for series:

Sub-case 1 $m > 0$:

$$T(n) = \Theta\left(n^{\log_b a} (L_k(n))^m\right).$$

Sub-case 2 $m = 0$:

$$T(n) = \Theta\left(n^{\log_b a} L_{k-1}(n)\right).$$

Sub-case 3 $m = 0$:

$$T(n) = \Theta\left(n^{\log_b a}\right).$$

Case 3: If $f(n) = \Omega(n^c)$ for some $c > \log_b a$, and the regularity condition that $\exists m \in \mathbb{R}, 0 < k < 1$ such that $n \geq m$ implies

$$af\left(\frac{n}{b}\right) \leq kf(n)$$

is satisfied, then

$$af\left(\frac{n}{b}\right) \leq kf(n)$$

$$a^j f\left(\frac{n}{b^j}\right) \leq a^{j-1} kf\left(\frac{n}{b^{j-1}}\right) \leq k^j f(n)$$

$$T(n) = \sum_{j=0}^{h-1} a^j f\left(\frac{n}{b^j}\right) + n^s T(1)$$

$$= f(n) + \sum_{j=1}^{h-1} O(k^j f(n)) + \Theta(n^s)$$

$$\sum_{j=1}^{\infty} k^j f(n) = \frac{f(n)}{1-k}$$

$$T(n) = f(n) + O(f(n)) + \Theta(n^s) = \Theta(f(n))$$

□

Terms:

- h is called recursion level.
- $\log_b a$ is called critical exponent c_{crit} .
- $T(1)$ is called leaf cost.
- At each node, the first term is called recursive work, and the second term is called non-recursive work.

For original master theorem, the second case is

1. If for some positive integers m ,

$$f(n) = \Theta\left(n^{\log_b a} (\ln n)^{m-1}\right),$$

then

$$T(n) = \Theta\left(n^{\log_b a} (\ln n)^m\right).$$

2. If

$$f(n) = o\left(n^{\log_b a} (\ln n)^{-1}\right),$$

then

$$T(n) = \Theta\left(n^{\log_b a} \ln(\ln n)\right).$$

3. If for some positive integers m ,

$$f(n) = \Theta\left(n^{\log_b a} (\ln n)^{-m-1}\right),$$

then

$$T(n) = \Theta\left(n^{\log_b a}\right).$$